

Sheet 10

Monday 15th December, 2025

Please hand in a zip-archive, containing a pdf for exercise 1, 2 and 3, as well as the plots and input files for exercise 2 and 3, which is named $\langle \text{firstname} \rangle_ \langle \text{lastname} \rangle_ \text{sheet}_ \langle \text{number} \rangle$.

The deadline is Monday, Dec. 22nd, 12:00 (noon). Please hand in your solution via moodle.

1 The Nose thermostat

When we perform a molecular dynamics simulation, the most natural choice of ensemble is microcanonical, so the number of particles N , the volume V and the energy E are all conserved. However, in physical systems, we usually want to fix the temperature T , not the total energy. In order to simulate this, we add a virtual variable s for the heatbath, such that performing a microcanonical simulation for the virtual system will result in a canonical ensemble for the full system.

We start with the real system, with coordinates $\mathbf{r}_i$ ($\mathbf{r}$ will refer to the vector of all coordinates) and momenta $\mathbf{p}_i$. The Hamiltonian for this system is then given by

$$H_{\text{real}} = \sum_{i=1}^N \frac{1}{2m_i} \mathbf{p}_i^2 + U(\mathbf{r}). \quad (1)$$

We introduce the bath degree of freedom s with its conjugate momentum p_s and mass Q , and define the Nose Hamiltonian, given by

$$H_{\text{Nose}} = \sum_{i=1}^N \frac{1}{2m_i} \mathbf{p}_i^2 + U(\mathbf{r}) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s). \quad (2)$$

Here we defined the potential for the bath as $\frac{3N+1}{\beta} \ln(s)$, where $\beta = \frac{1}{k_B T}$ is determined by the temperature of the system. The mass Q has to be chosen by the user, and will determine how well the method converges in practice, but its value is not relevant to this discussion.

1. Show that performing the transformation to the virtual Nose-system $\mathbf{r}_N = \mathbf{r}$, $\mathbf{p}_N = s\mathbf{p}$, $dt_N = sdt$ results in the Hamiltonian

$$H_{\text{Nose}} = \sum_{i=1}^N \frac{1}{2m_i s^2} \mathbf{p}_{N,i}^2 + U(\mathbf{r}_N) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s). \quad (3)$$

1 point

2. Show that calculating the partition function for the microcanonical ensemble of the Nose-system can be rewritten as calculating it for the canonical ensemble of the real system (up to a constant C), by transforming back to the real coordinates and integrating out the heat bath degrees of freedom s, p_s .

$$Z_{\text{Nose}}^{\text{micro}} = \int d^{3N} p_N d^{3N} r_N ds dp_s \delta(H_{\text{nose}} - E) = C \int d^{3N} p d^{3N} r e^{-\beta H_{\text{real}}} = C \cdot Z_{\text{real}}^{\text{can}}. \quad (4)$$

How can we use expectation values calculated with the microcanonical ensemble for the virtual system to get expectation values calculated with the canonical ensemble for the real system?

Hint: Insert an observable $O(\mathbf{r}, \mathbf{p})$ into the definition of the canonical expectation value, and perform the derivation for eq. 4 backwards.

2 points

2 NVT simulations of Si

Unless stated differently, you can use a total simulation time of 10 ps. Use the input files from the previous sheet. Please increase the system size to 512 atoms.

Hints: You can use the replicate command in lammps to increase the system size. If you run lammps in parallel (see listing 1), you can speed up calculations. You do not have to save trajectories for the following tasks.

Listing 1: Command to run lammps on two cores in parallel.

```
mpiexec -n 2 lammps < input > output
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1. Change the input such that a calculation at $T = 300$ K is performed, instead of a calculation with constant energy
Hint: use fix NVT ...
2. Plot volume, pressure, temperature, and total energies as functions of time. What do you observe for the temperature?
Please use a different plot for each quantity and provide units.
3. Calculate the average values and standard deviations of the above observables.

3 points

3 NPT simulations of Si

1. Change the input such that a calculation at $T = 300$ K and $p = 0$ is performed.
Hint: use fix NPT ...

2. Plot volume, pressure, temperature, and total energies as functions of time. What do you observe for pressure and volume?
Please use a different plot for each quantity and provide units.
3. Calculate the average values and standard deviations of the above observables.
4. For a system of 512 atoms, change the temperature from 100 K to 1000 K in steps of 100 K and plot the average volume as a function of average temperature. What do you observe? Can you give a physical interpretation?
Hint: calculate the average pressure and temperature after the system is equilibrated.

4 points